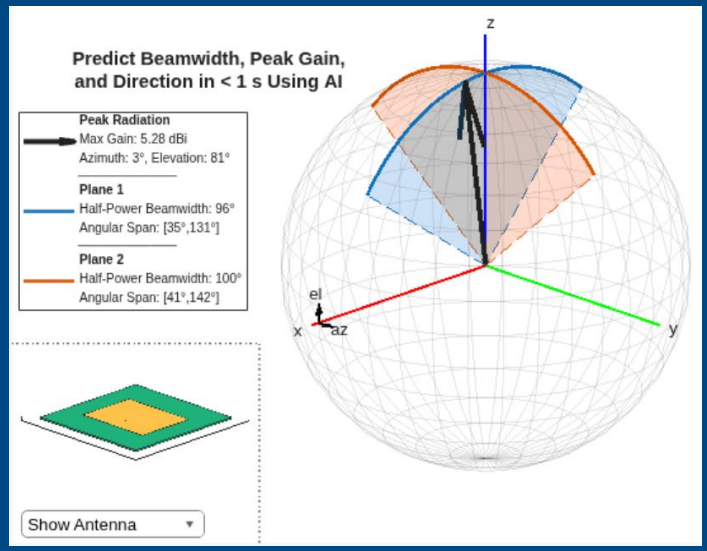
# MILCOM 2027 — Tutorial TUT13
# AI-Enabled Antenna-to-System Design with LLMs, MCP and MATLAB

*Part 2 — AI Models and Pattern Reconstruction*

*2A — Pretrained Antenna Models and AutoML*

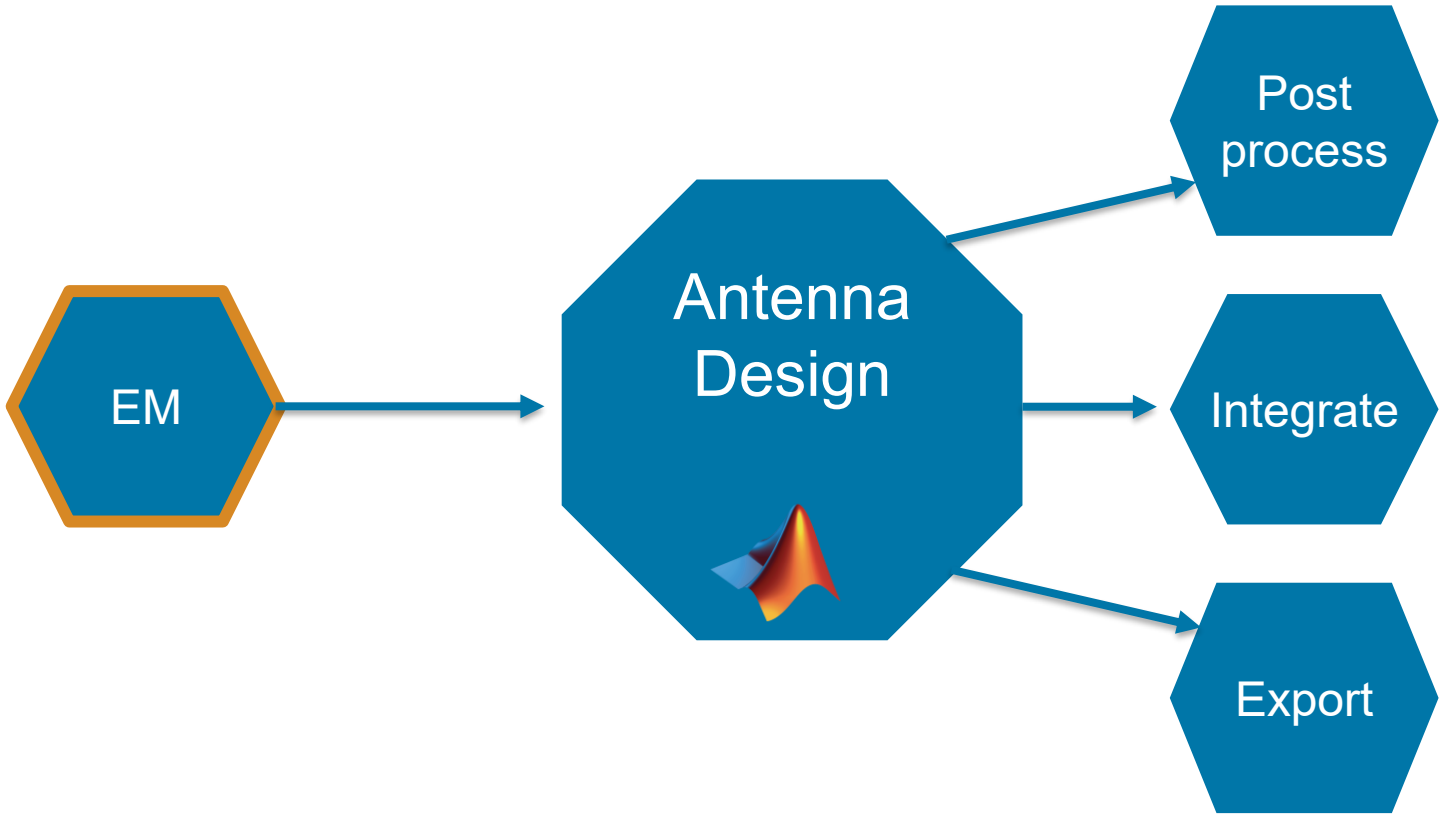
Sudarshan Sivaramakrishnan – ssivaram@mathworks.com

*Principal Software Engineer*
*MathWorks | RF/EM Group*




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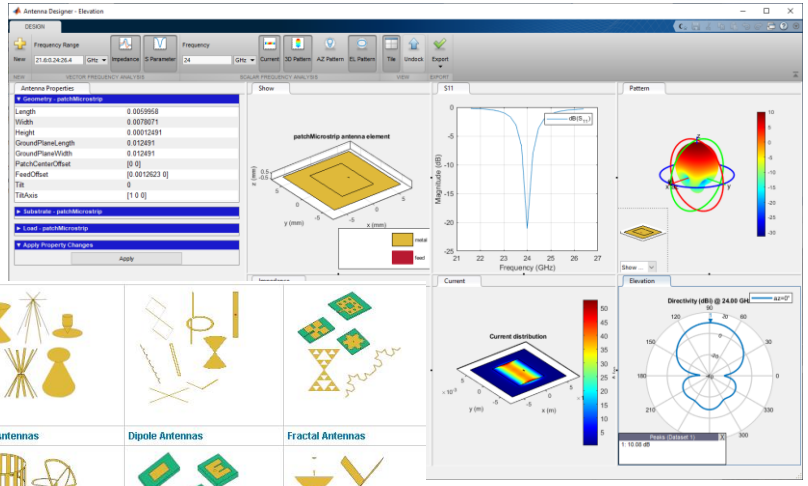
# Antenna Workflow #1: Electromagnetic Analysis



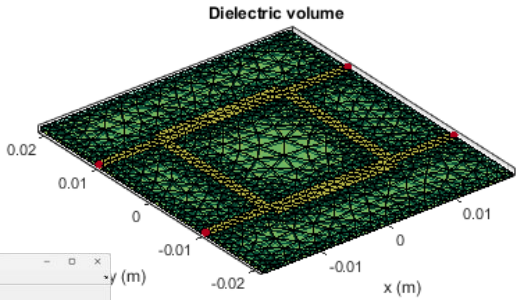
# Antenna Toolbox

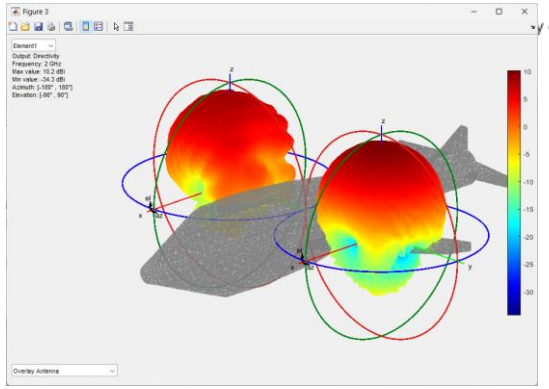
## Design, analyze, and visualize antenna elements and antenna arrays

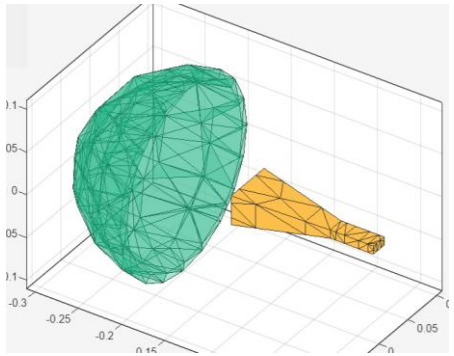
- Get started with antenna and array catalog and apps
- Perform full-wave electromagnetic analysis (MoM, FMM, …)
- Design custom planar and 3D structures
- Integrate with RF PCB Toolbox for board design (FEM)
- Analyze the effects of installation on large platforms
- Include RF propagation effects

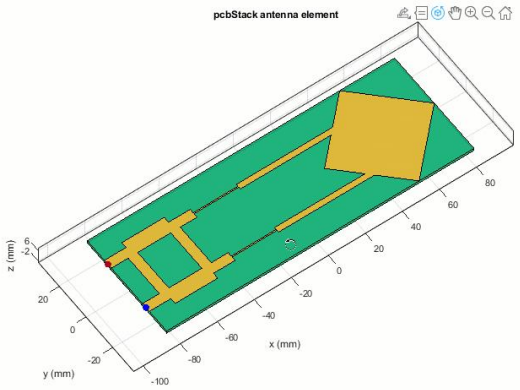




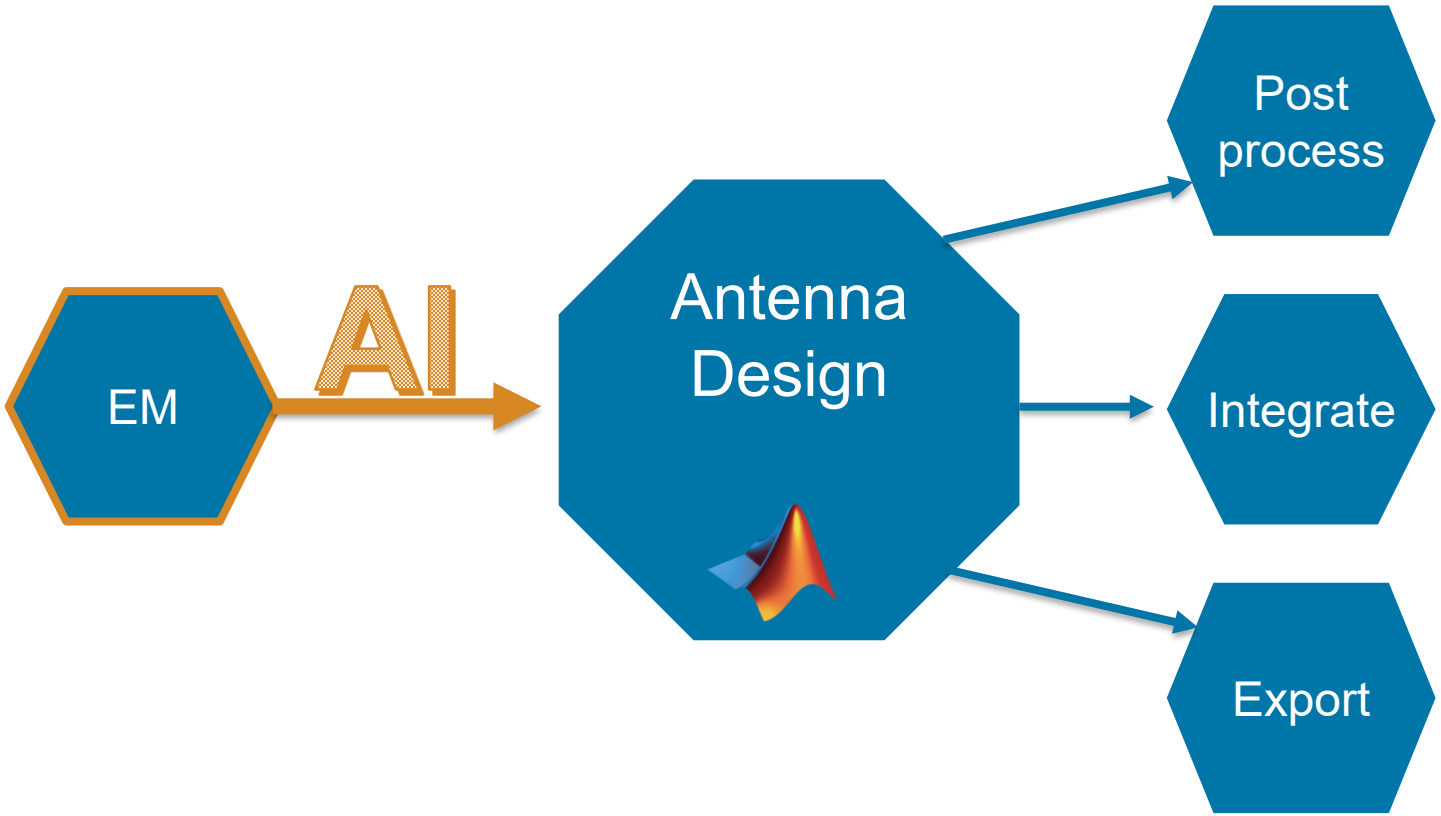







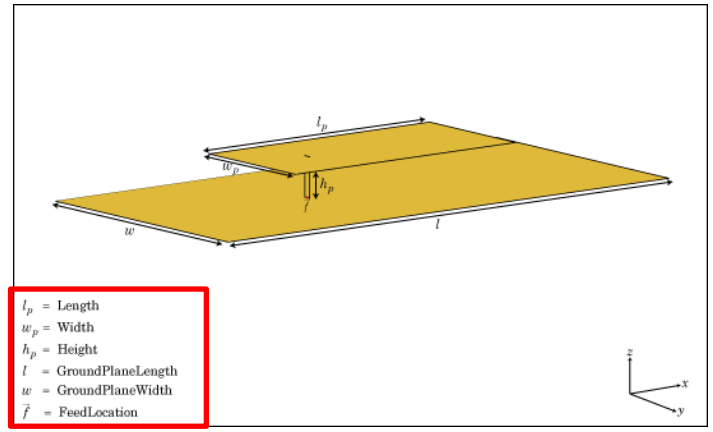
# Antenna Workflow #1: Electromagnetic Analysis with AI

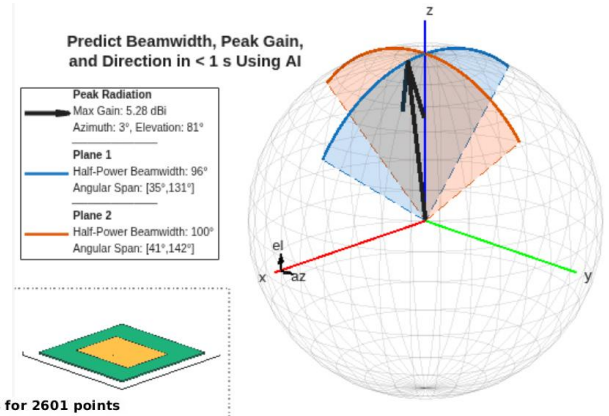


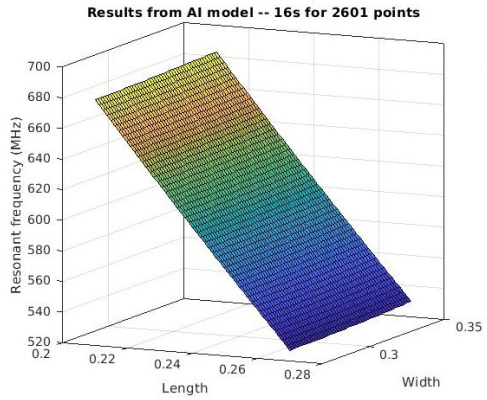

- Do you need to explore large design spaces?
- Do you want to speed up antenna analysis with AI?

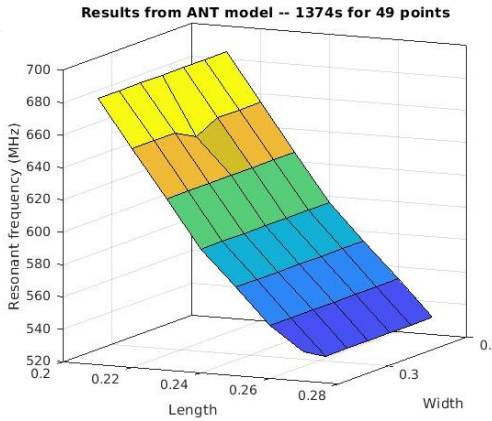
# Pretrained AI Antenna Models

- Unique solution within a systematic and extensible framework
- Based on subset of antenna catalog
  - Ready-to-use models, no AI expertise needed
  - Preselected tunable parameters
  - Dipole, dipoleHelix, horn, waveguide, patch antennas (8 types)
  - Substrate materials from dielectric catalog
    (supported for 4 antenna types now, and the list is growing)
- Enable AI-based analysis
  - Rapid characterization
  - Design space exploration
  - Port analyses: resonant frequency, bandwidth
  - Field analyses: beamwidth, peak radiation




https://www.mathworks.com/help/antenna/ref/em.ai.aiantenna.html

https://www.mathworks.com/help/antenna/ug/ai-based-design-space-exp-analysis-and-optimization.html

S. Sivaramakrishnan, V. Iyer, T. Gao and G. Zucchelli, "A Systematic Application of AI Techniques to Antenna Design, Analysis, and Optimization," *2024 54th European Microwave Conference (EuMC)*, Paris, France, 2024, pp. 972-975, doi: 10.23919/EuMC61614.2024.10732562

# Rapid Design Exploration Using Pretrained AI Models

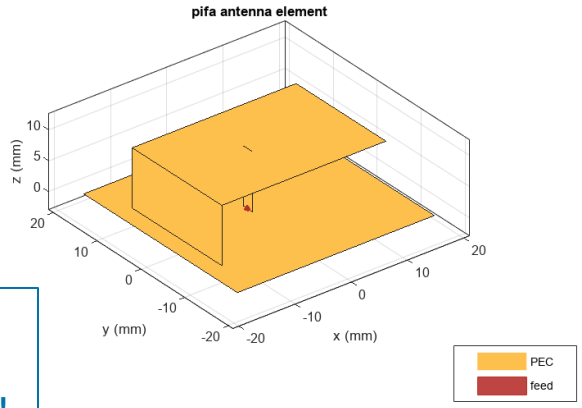
```
>> p = design(pifa,2.4e9,ForAI=true)
p =
  AIAntenna with properties:

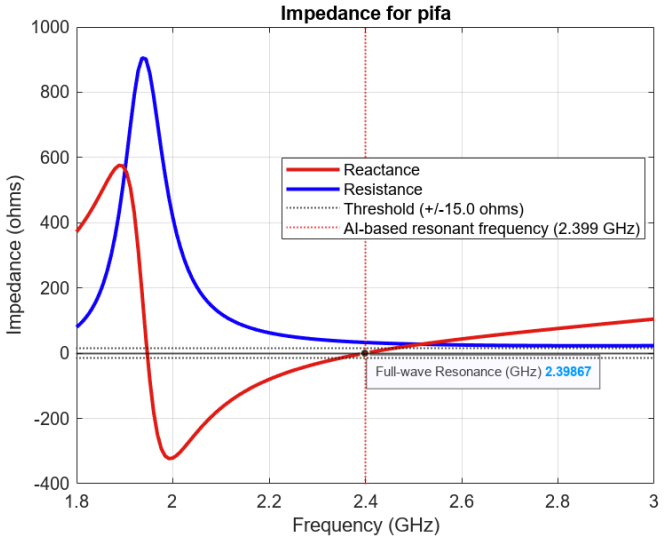
   Antenna Info
               AntennaType: 'pifa'
    InitialDesignFrequency: 2.4000e+09
   Tunable Parameters
                    Length: 0.0292
                     Width: 0.0194
                    Height: 0.0097
             ShortPinWidth: 0.0194
Show read-only properties
```

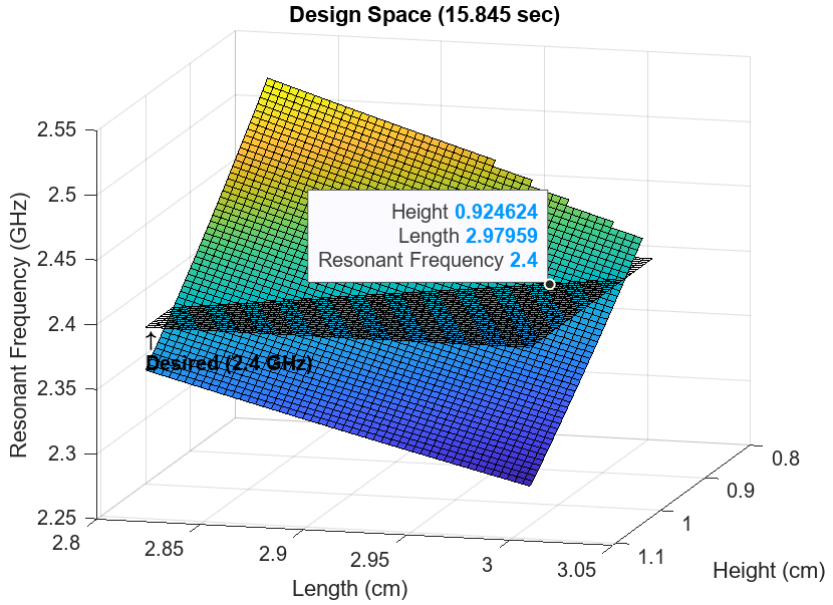
Iteratively repeat this after modifying tunable parameters!

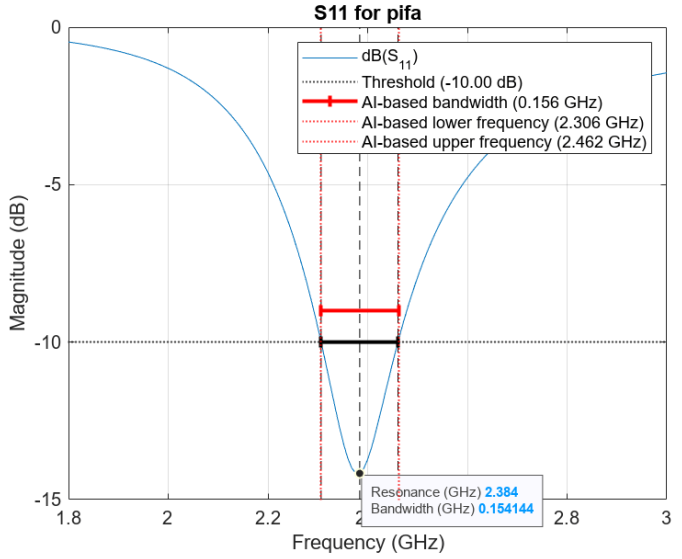
```
>> tic, fR = resonantFrequency(p), toc
fR =
   2.4091e+09
Elapsed time is 0.158043 seconds.
```

```
>> tic, [absBW,fL,fU,matching] = bandwidth(p), toc
absBW =
   1.4809e+08
fL =
   2.3186e+09
fU =
   2.4667e+09
matching =
  categorical
     Matched
Elapsed time is 0.190911 seconds.
```



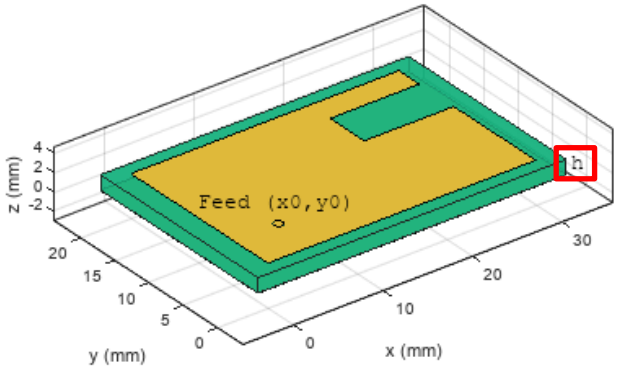

Artificial Intelligence (AI) for Rapid Analysis and Design of Microstrip Patch Antenna
https://www.mathworks.com/help/antenna/ug/ai-for-rapid-analysis-and-design-of-microstrip-patch.html

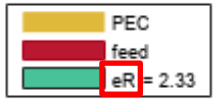
# Interactive Exercises Using MATLAB Online R2026a

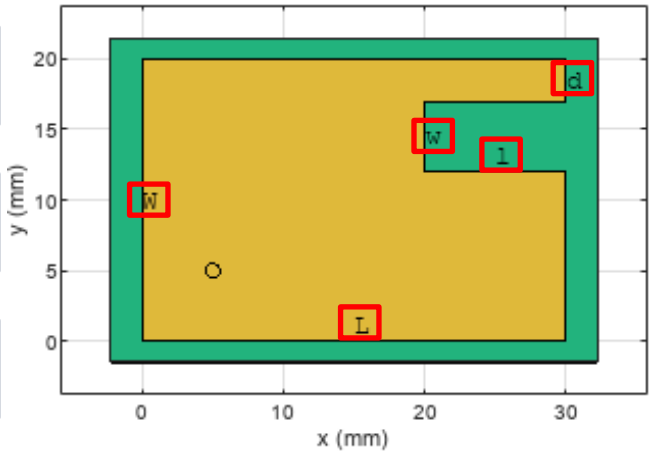
# Live Demo Using R2026b Prerelease

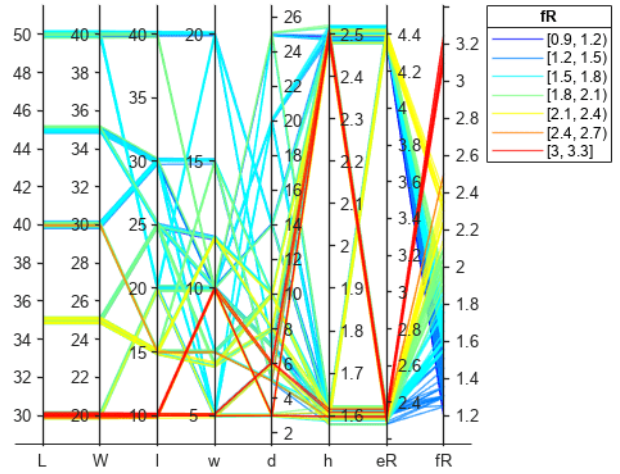
# Train your AI Antenna Model

- **Specify** parameterized antenna structure
- **Identify** predictors and responses
- **DOE** (design of experiments)
- **Generate** data using full-wave EM simulation
- **Prepare** and preprocess data for training
- **Train** machine learning model (use AutoML)
- **Evaluate** ML model vs EM simulation



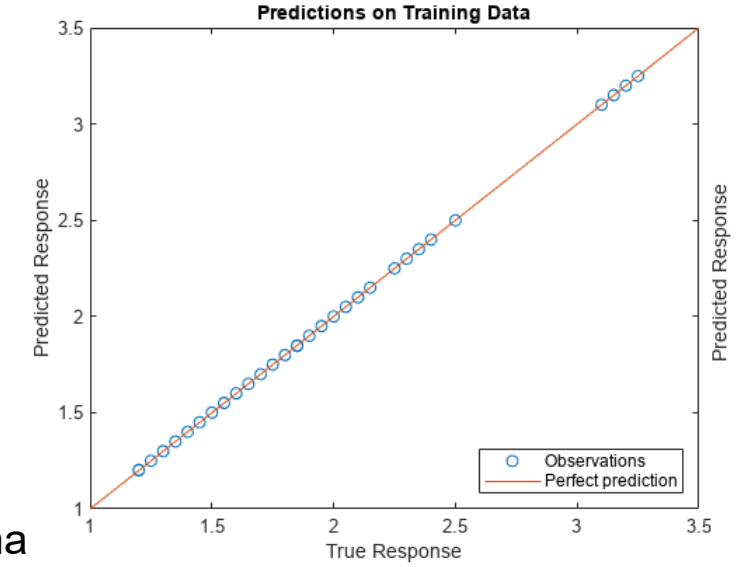



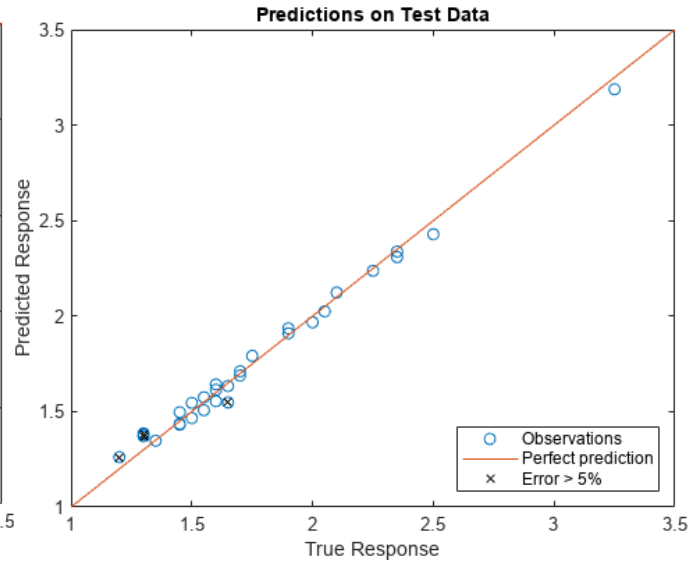




```
Optimization completed.
Total iterations: 60
Total elapsed time: 370.6171 seconds
Total time for training and validation: 252.4126 seconds

Best observed learner is a gp model with:
        Learner:                        gp
        Sigma:                  0.0073413
        BasisFunction:                none
        KernelFunction: ardexponential
        KernelScale:                  NaN
        Standardize:                false
Observed log(1 + valLoss): 0.023284
Time for training and validation: 1.2777 seconds

Best estimated learner (returned model) is a gp model with:
        Learner:                        gp
        Sigma:                  0.0073413
        BasisFunction:                none
        KernelFunction: ardexponential
        KernelScale:                  NaN
        Standardize:                false
Estimated log(1 + valLoss): 0.023767
Estimated time for training and validation: 1.3343 seconds
```

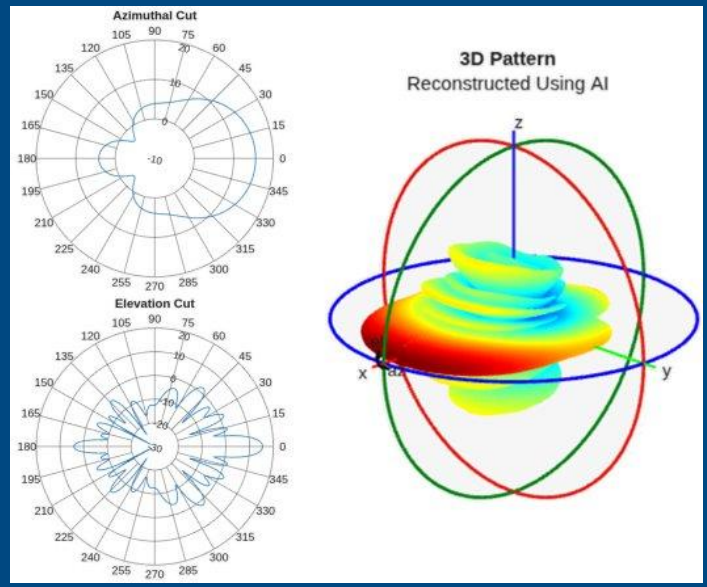



Train Machine Learning Model for Analysis of Custom Antenna
https://www.mathworks.com/help/antenna/ug/train-mlmodel-for-analysis-of-custom-antenna.html

# MILCOM 2027 — Tutorial TUT13
# AI-Enabled Antenna-to-System Design with LLMs, MCP and MATLAB

*Part 2B — AI-Based Pattern Reconstruction*

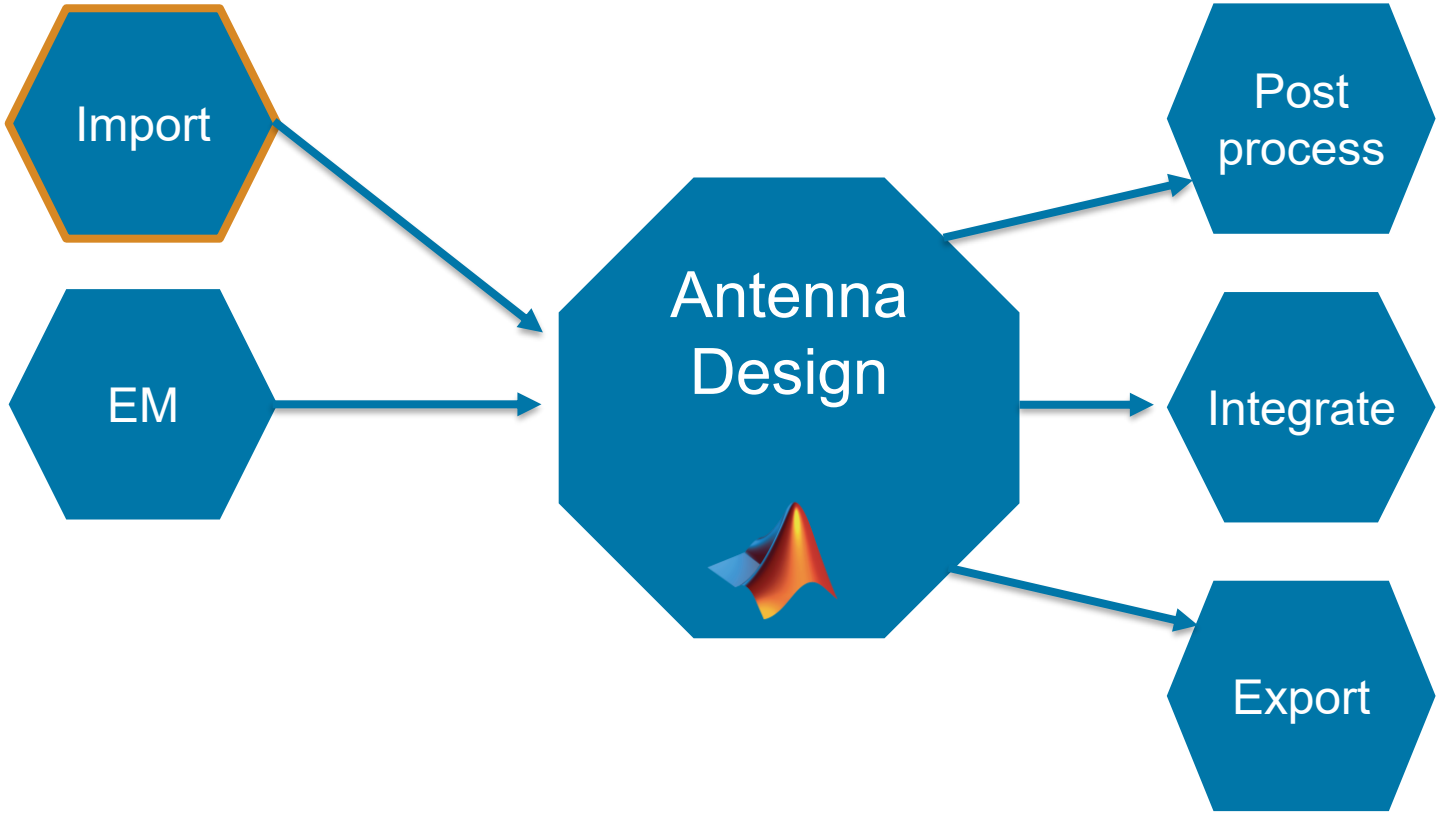



Sudarshan Sivaramakrishnan – ssivaram@mathworks.com

*Principal Software Engineer*
*MathWorks | RF/EM Group*

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# Antenna Workflows #2: Importing Data



# Importing Patterns – File Readers

- File Formats
  - .CSV
  - .MSI, .PLN
  - .FFS (CST)
  - .FFD (Ansys)
- Query on struct/object
- Visualize using
  - **`polarpattern`**
  - **`patternCustom`**

```
>> [Horizontal,Vertical,Optional] = msiread('Sample.pln')

Horizontal = struct with fields:
    PhysicalQuantity: 'Gain'
           Magnitude: [360x1 double]
               Units: 'dBd'
             Azimuth: [360x1 double]
           Elevation: 0
           Frequency: 659000000
               Slice: 'Elevation'

Vertical = struct with fields:
    PhysicalQuantity: 'Gain'
           Magnitude: [360x1 double
               Units: 'dBd'
             Azimuth: 0
           Elevation: [360x1 double
           Frequency: 659000000
               Slice: 'Azimuth'

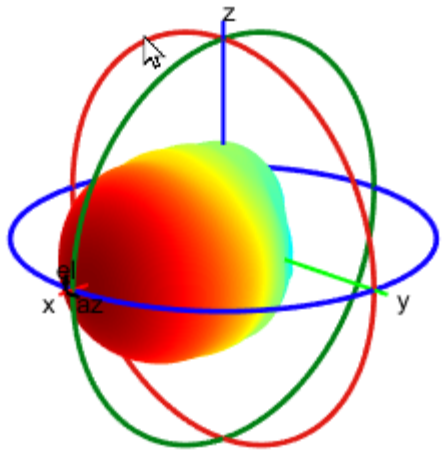
Optional = struct with fields:
            name: 'Sample.pln'
            make: 'Sample 4DR-16-2HW'
       frequency: 659000000
         h_width: 180
         v_width: 7.3000
    front_to_back: 34
            gain: [1x1 struct]
            tilt: 'MECHANICAL'
    polarization: 'POL_H'
         comment: 'Ch-45 0 deg dt'
    scaling_mode: 'AUTOMATIC'
```

Command Window

```
>> data = ffsread("C:\temp\test2.ffs")
data =

  struct with fields:

           Version: "3.0"
            Format: "Farfield"
     NumFrequencies: 1
          Alignment: [1×1 struct]
      RadiatedPower: 0.4863
      AcceptedPower: 0.4969
    StimulatedPower: 0.5000
          Frequency: 3.3000e+11
                Phi: [65341×1 double]
              Theta: [65341×1 double]
         EFieldTheta: [65341×1 double]
           EFieldPhi: [65341×1 double]
```



# Using Antenna Data: Introducing the Measured Antenna Object

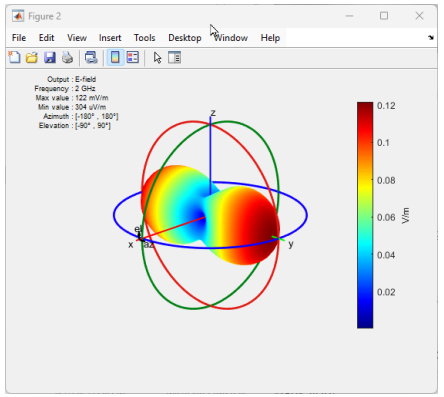
- Import antenna data: from measurements or 3rd party EM tools
  - S-parameters
  - E-field of (embedded) element

```
>> pattern(mesAnt,fRange(2),Type="efield")
>> EHfields(mesAnt,fRange(2))
>> sp = sparameters(mesAnt,fRange);
```

```
mesAnt = measuredAntenna(E=eField, Direction=dir, NumPorts=numAnt,...
                Azimuth=az, Elevation=el, FieldCoordinate="polar",...
                EmbeddedE=embE, FieldFrequency=fRange, Sparameters=s)
```

```
mesAnt =
  measuredAntenna with properties:

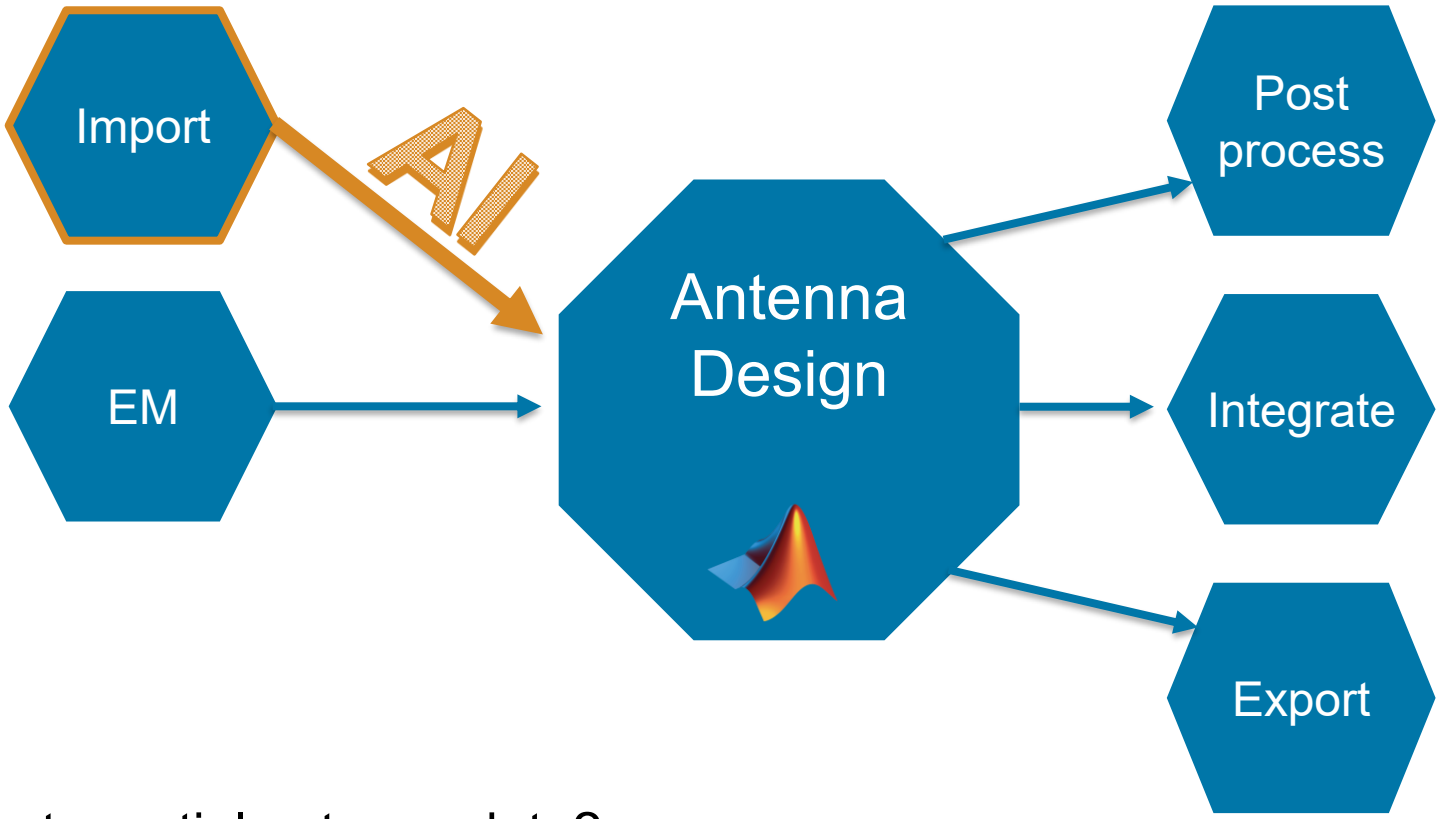
                    E: [2701x3x3 double]
            Direction: [2701x3 double]
          PhaseCenter: [0 0 0.0750]
             NumPorts: 2
        FieldFrequency: [3x1 double]
       FieldCoordinate: "polar"
              Azimuth: [-180 -175 -170 -165 -160 -155 -150
            Elevation: [180 175 170 165 160 155 150 145 14
         Sparameters: [1x1 sparameters]
       AmplitudeTaper: 1
           PhaseShift: 0
            EmbeddedE: [2701x3x2x3 double]
   TerminationImpedance: 50
    CalculateTotalField: 0
```




**`Direction`**: No. of observation points (Px3)
**`FieldFrequency`**: No. of frequency points (F)
**`NumPorts`**: Number of feeds (N)

**`E`**: P x 3 x F
**`EmbeddedE`**: P x 3 x N x F
**`Sparameters`**: N-port s-parameters object

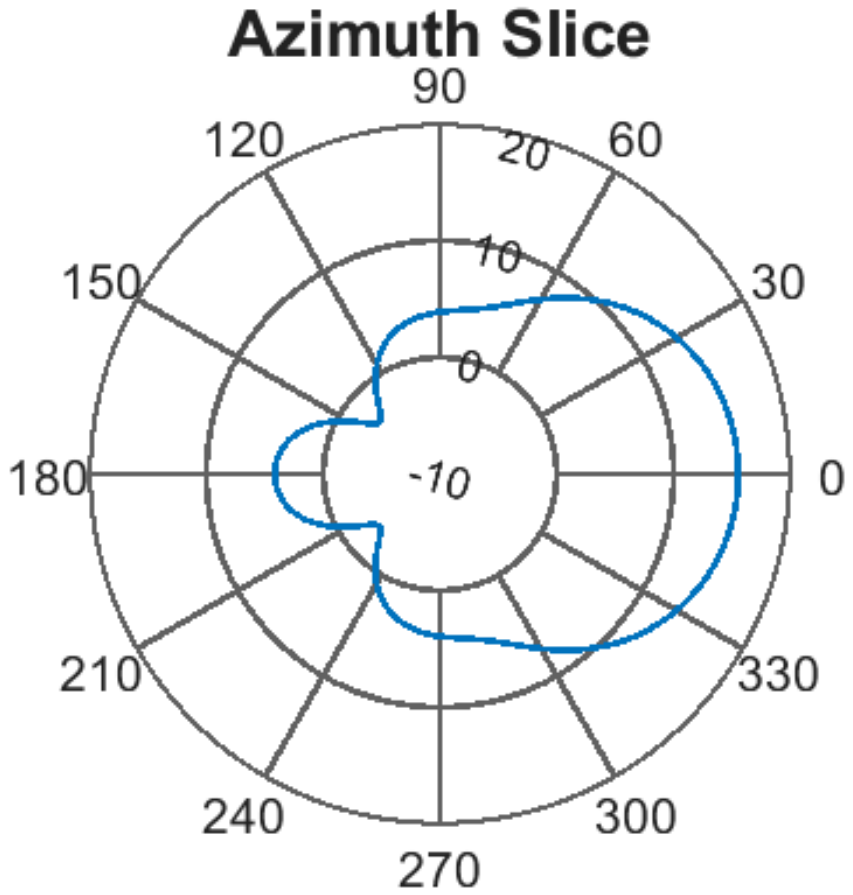
# Antenna Workflows #2: Importing Data with AI

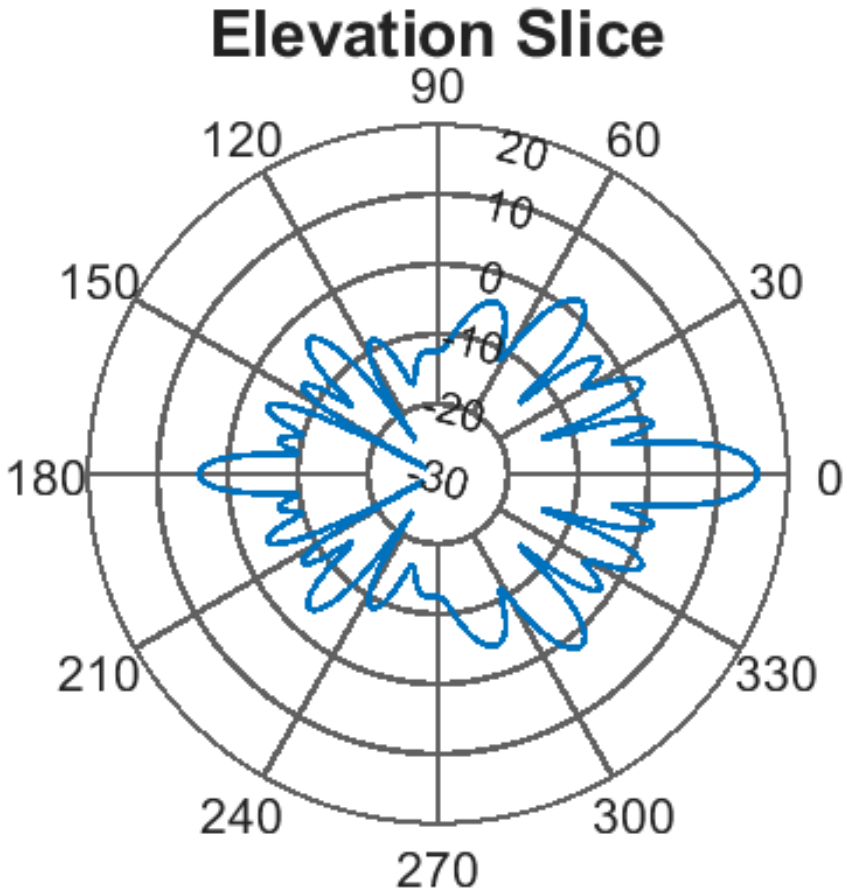



- Do you have access to partial antenna data?
- Do you need to integrate antennas at the system-level?
- Do you need to speed up antenna measurements?

# What if my Pattern is not Complete?

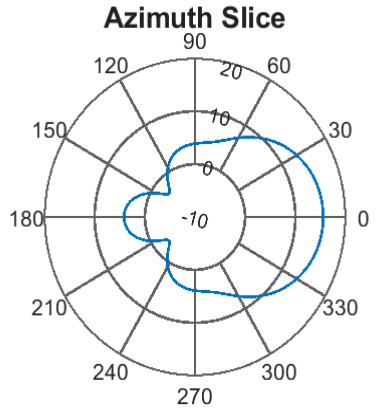
*Can you reconstruct the full (3D) pattern?*

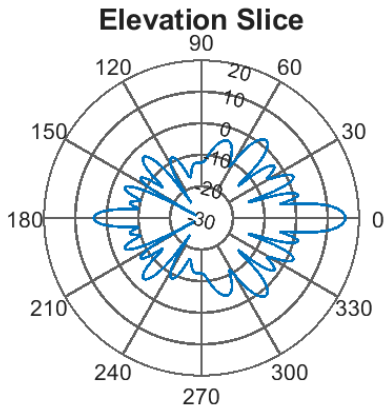


# What if my Pattern is not Complete?

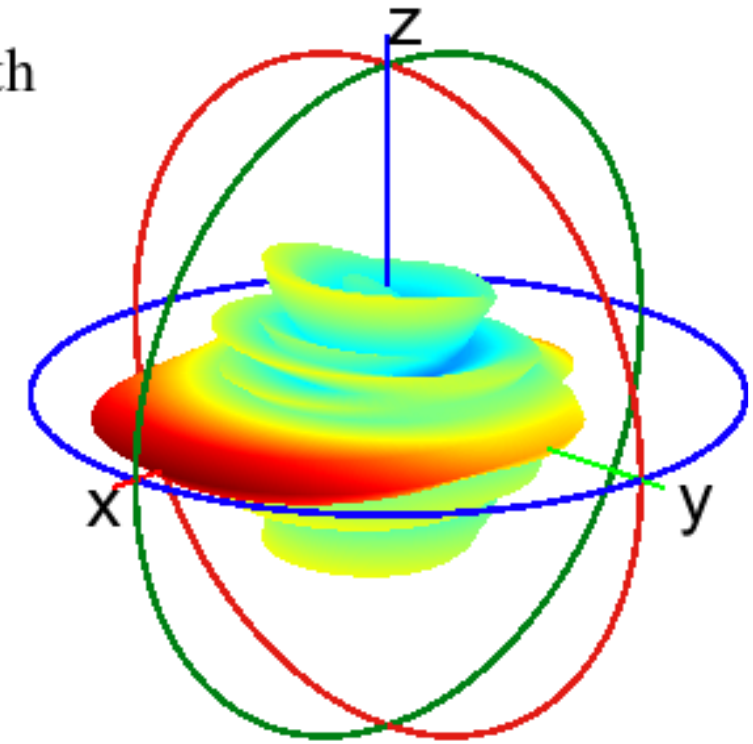
- Motivation for reconstructing 3D radiation patterns:
  - Enable system designers working with antenna datasheets (partial data)
  - Reduce the number of measurement samples



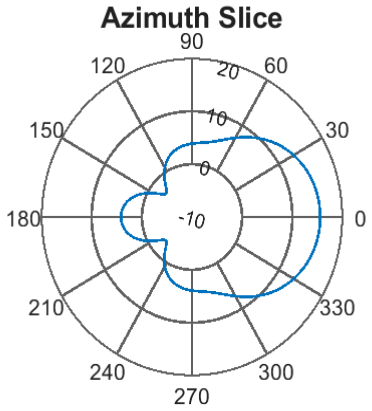

?

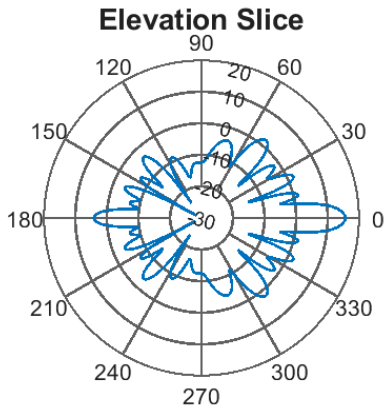
Ground Truth



# What if my Pattern is not Complete? 3D Pattern Reconstruction

- Reconstruct patterns starting from 2 (orthogonal) slices:
  - Analytical methods: **`patternFromSlices`**
  - Deep learning based prediction: **`patternFromAI`**



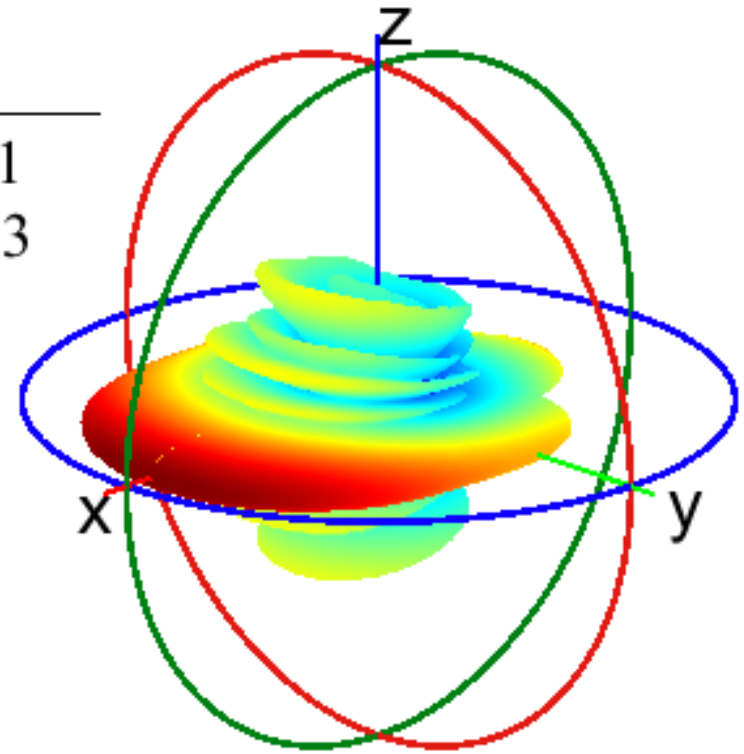




3D
Pattern Reconstruction

AI-Based

RMSE = 7.31

SSIM = 0.653




https://www.mathworks.com/help/antenna/ref/patternfromai.html
https://www.mathworks.com/help/antenna/ref/patternfromslices.html

# AI-Enabled Pattern Reconstruction – *Overview*

## *Why AI?*

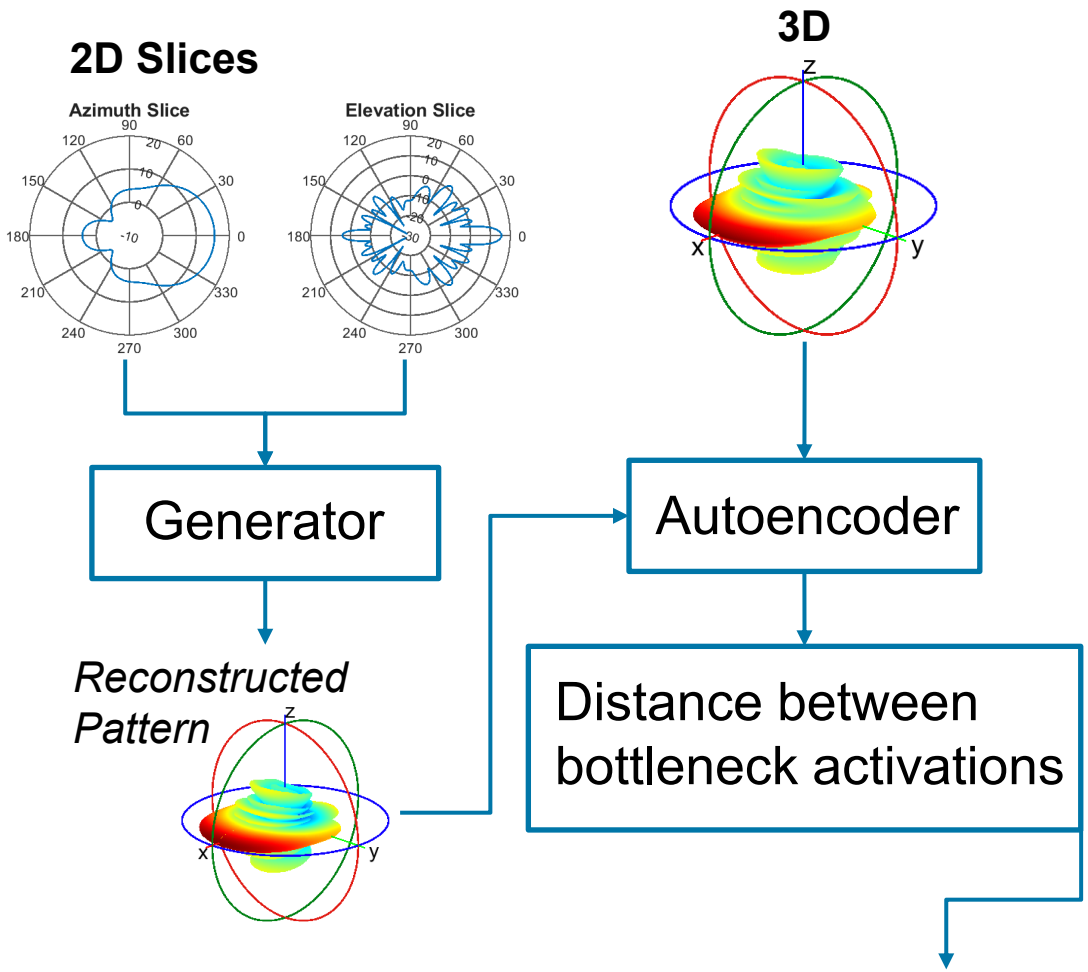
- Improve the accuracy of the reconstructed 3D pattern
- Overcome limitations of analytical approaches with no assumptions on input data format

## *Training Process*

- Generator learns to predict 3D pattern given slices
- Custom loss function to compare against ground truth
  - L1: mean absolute error
  - SSIM: structural similarity
  - Feature-based loss

## *Training data: ~100,000 unique patterns (EM simulation)*

- Single port antennas: omni / semi / highly-directional
- Array configuration: linear, circular, rectangular
- Classic image augmentations: noise, transformations



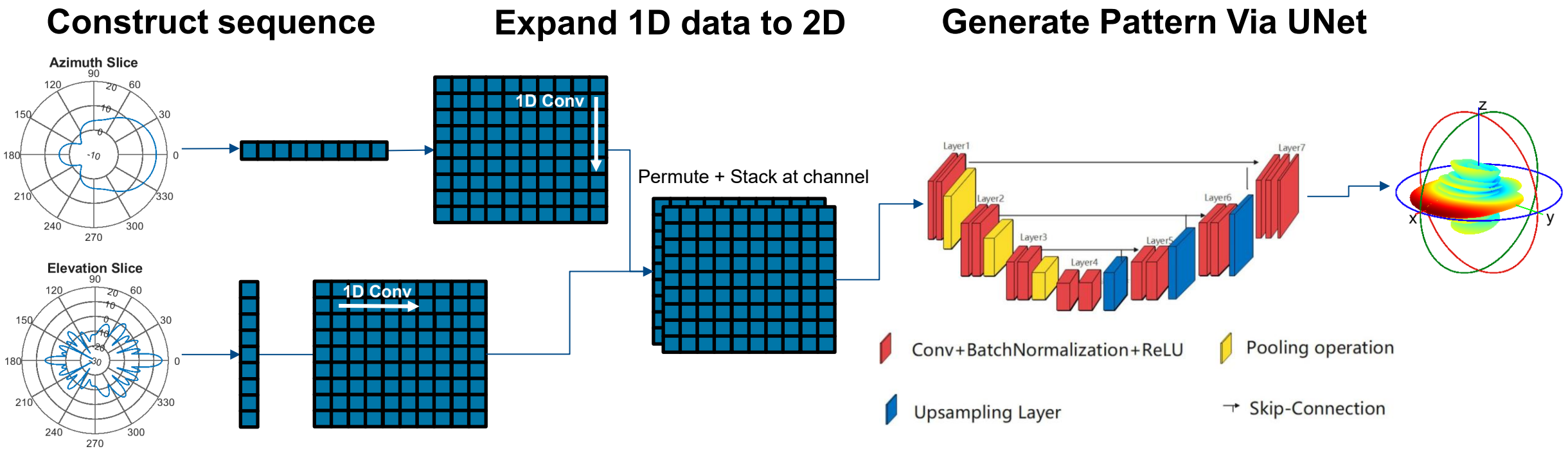

$$Total\ Loss = \alpha \cdot (L1\ Loss) + \beta \cdot (SSIM\ Loss) + \gamma \cdot (Feature\ Loss)$$

$\alpha, \beta, \gamma$ = hyper parameter weights


P. J. DiMeo, S. Sivaramakrishnan and V. Iyer, "AI-Based 3D Antenna Pattern Reconstruction," *2024 IEEE International Symposium on Antennas and Propagation and INC/USNC-URSI Radio Science Meeting (AP-S/INC-USNC-URSI)*, Firenze, Italy, 2024, pp. 1-2, doi: 10.1109/AP-S/INC-USNC-URSI52054.2024.10686954.

# AI-Enabled Pattern Reconstruction – *Generator Architecture*

Generative model similar to multi-sequence-to-image convolutional neural network (CNN) with a U-Net backbone [1]





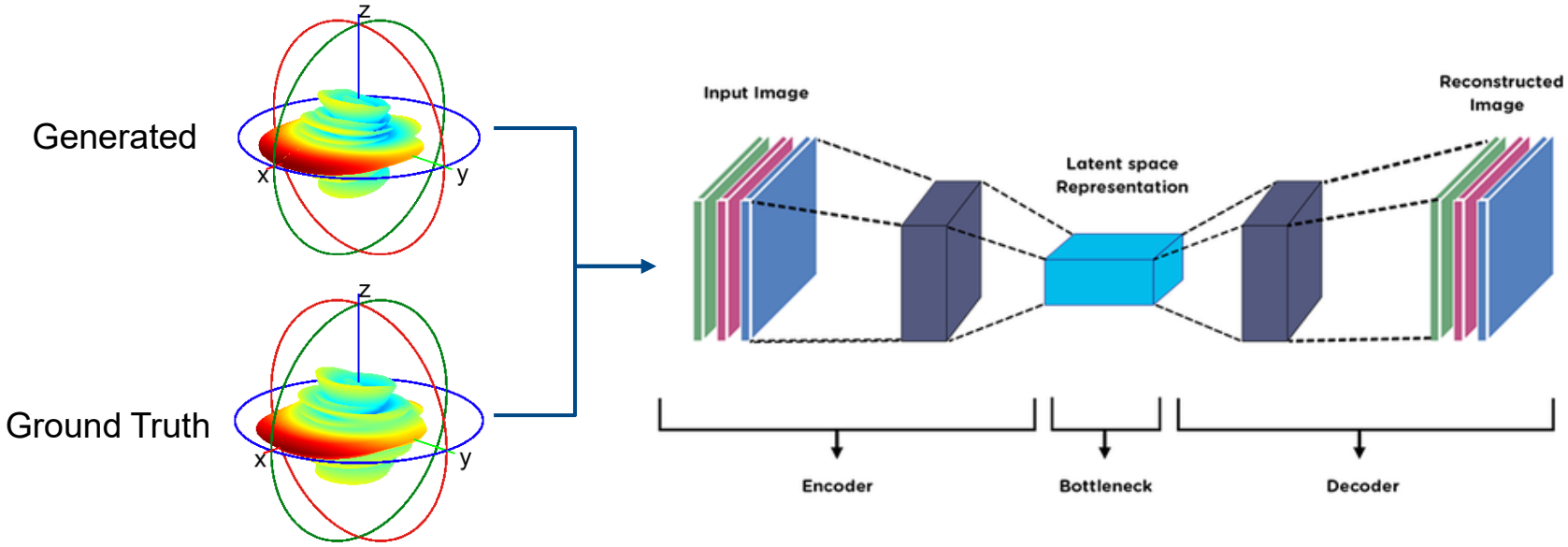
[1] V. Ghodrati, J. Shao, M. Bydder, Z. Zhou, W. Yin, K. L. Nguyen, Y. Yang, and P. Hu, "MR image reconstruction using deep learning: evaluation of network structure and loss functions," *Quant Imaging Med Surg.* vol. 9, no. 9, pp. 1516-1527, Sep. 2019. doi: 10.21037/qims.2019.08.10.

[2] S. Cai, Y. Wu, and G. Chen, "A Novel Elastomeric UNet for Medical Image Segmentation," *Front. Aging Neurosci.*, vol. 14, p. 841297, Mar. 2022. [Online]. Available: https://doi.org/10.3389/fnagi.2022.841297

# AI-Enabled Pattern Reconstruction – *Feature Loss Autoencoder*

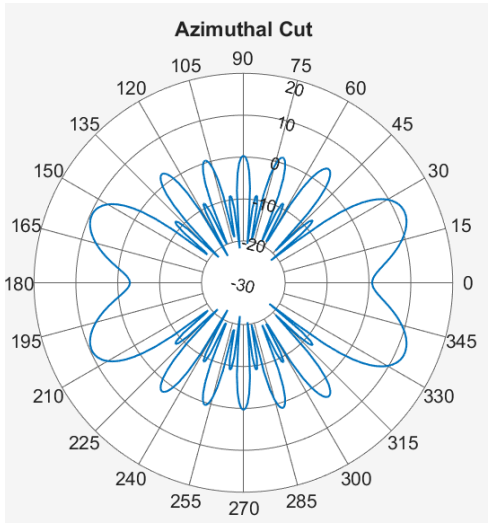
Custom-trained auto-encoder for (perceptual) feature-based loss

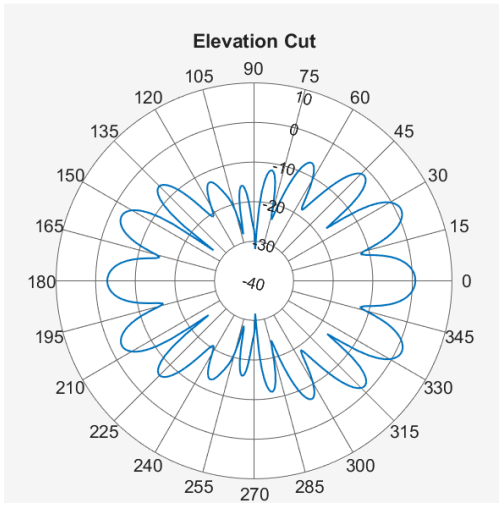
- Perceptual loss captures antenna specific features
- Bottleneck activations serve as the semantic representation
- Comparing activations at the **bottleneck** will guide us toward a more perceptually similar generated pattern.

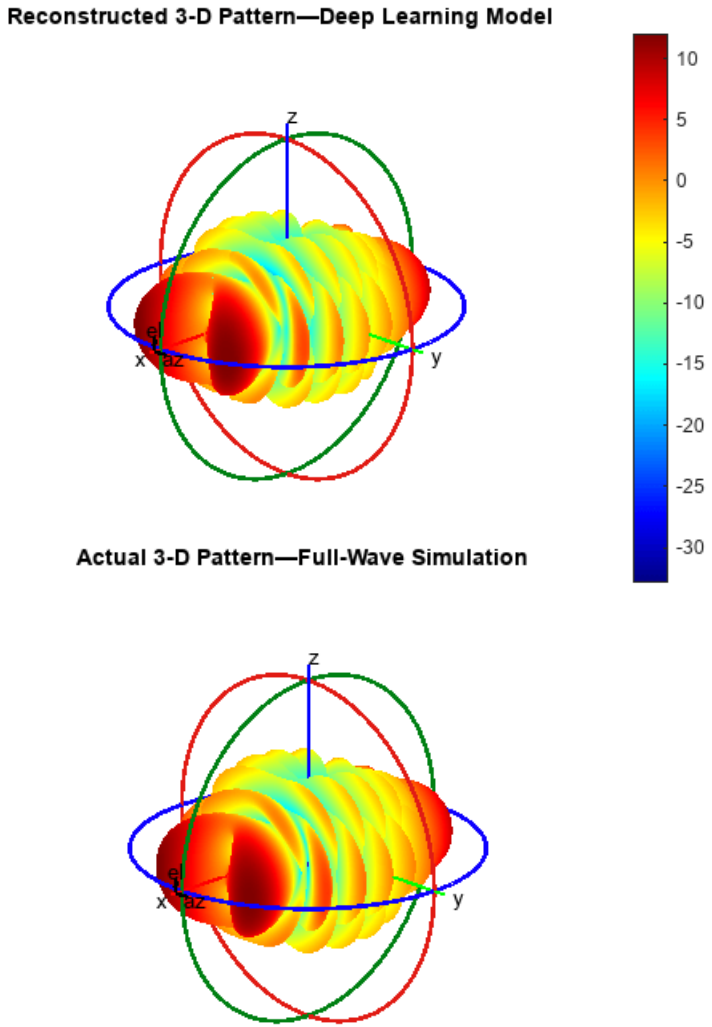


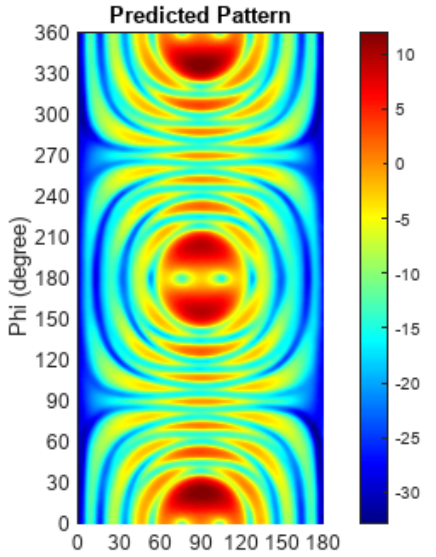

[1] https://medium.com/@birla.deepak26/autoencoders-76bb49ae6a8f

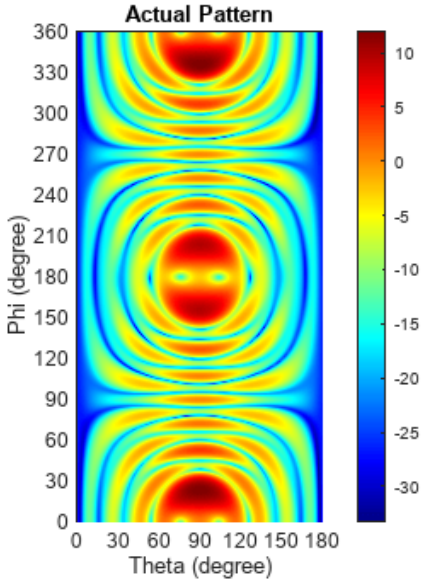
# Pattern Reconstruction Using AI: *Simulation Data Results*

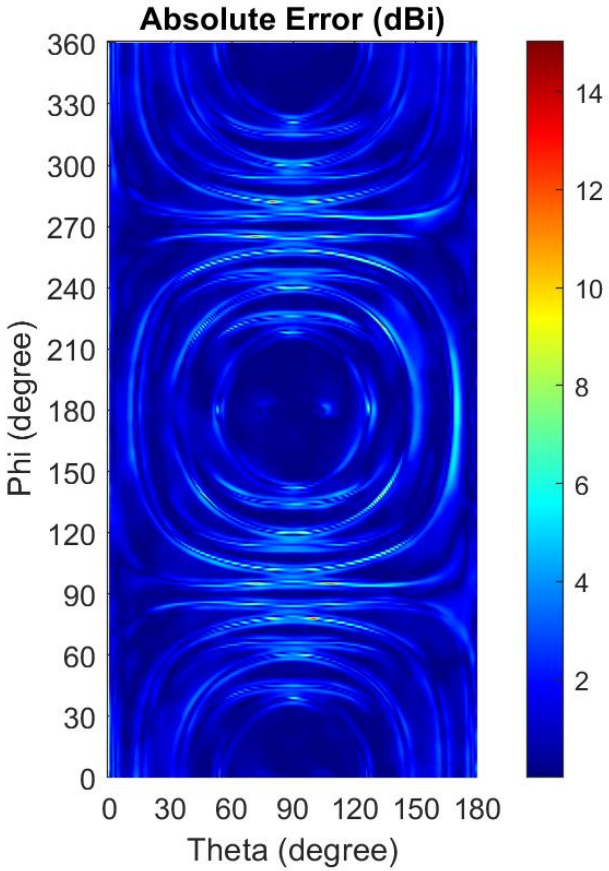


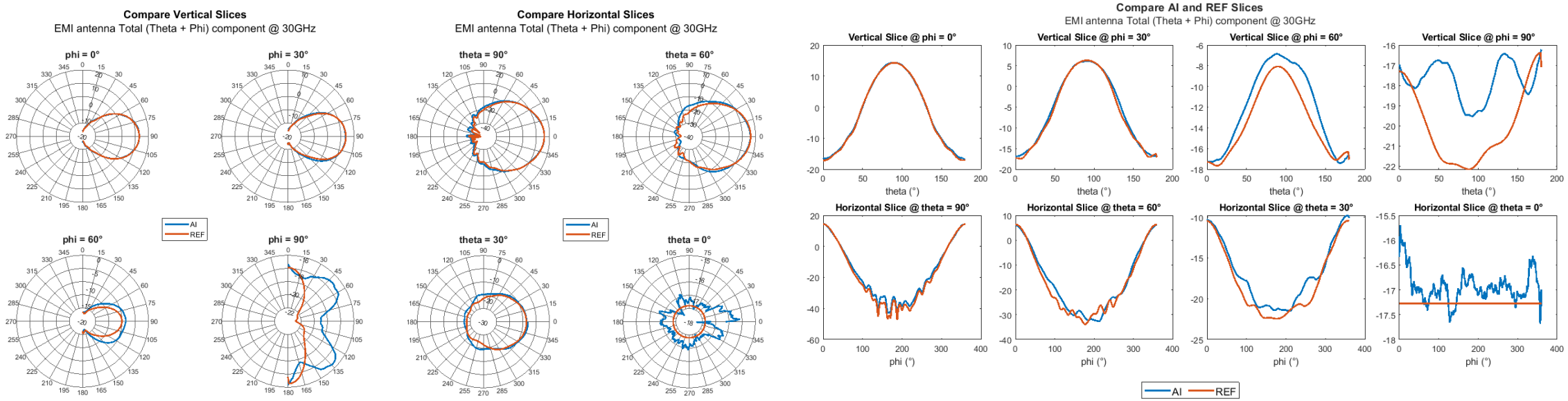

Reconstruct 3-D Antenna Pattern from 2-D Slices Using Deep Learning

https://www.mathworks.com/help/antenna/ug/reconstruct-3d-pattern-from-2d-slices-using-deep-learning.html

# Pattern Reconstruction Using AI: *Measured Data Results*

*Collaboration with Rohde&Schwarz*

- Unique commercial AI-based model to reconstruct diverse antenna pattern types
- Reconstruct 3D pattern from sparse data (~ **1%** of full pattern data) with low RMSE and high SSIM
- Small model size – **30 MB**
- Fast inference times of ~ <= **1sec**




Rohde&Schwarz Test Forum video: [Accelerating antenna pattern analysis and simulation with MATLAB]

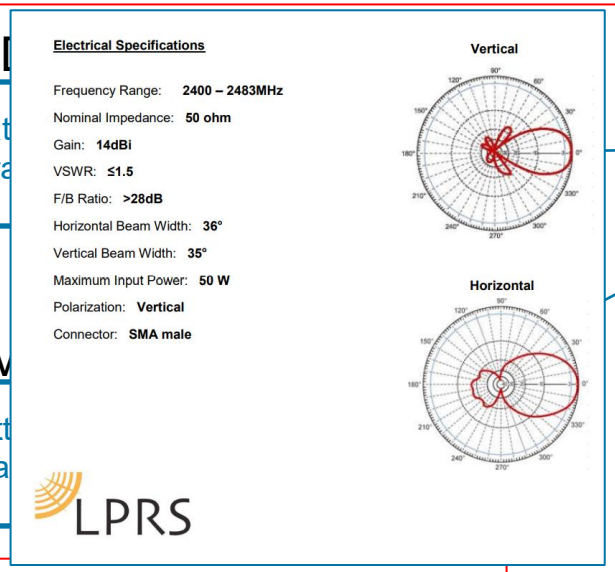
*With special thanks and acknowledgements to the work of Adam Tankielun (Rohde & Schwarz)*

# Interactive Exercises Using MATLAB Online R2026a

# Live Demo Using R2026b Prerelease

# Review and 3p integration workflow:

Convert image into 2D slice data (26a)

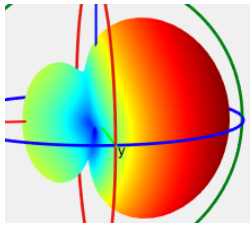



2d pat
S-para

2d patt
S-para

Read 3p file formats : ffs (25b)

3D pattern reconstruction



patternFromAI

3p simulations

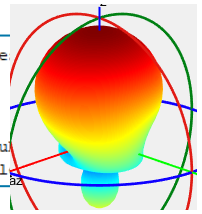
3d complex pattern data
S-parameter touchstone file

Measurement

3d complex pattern data
S-parameter touchstone file

sparameters with propertie

Impedance: 50
NumPorts: 1
Parameters: [1×1×31 dou
Frequencies: [31×1 doubl



Measured Antenna

measuredAntenna with properties:

E: [190×3×3 double]
Direction: [190×3 double]
PhaseCenter: [0 0 0]
NumPorts: 1
FieldFrequency: [3×1 double]
FieldCoordinate: 'polar'
Azimuth: [-180 -160 -140 -12
Elevation: [90 70 50 30 10 -1(
Sparameters: [1×1 sparameters]

System Simulation

PhasedArray

Ray Tracing

Satcom

Comms
Standards

Radar